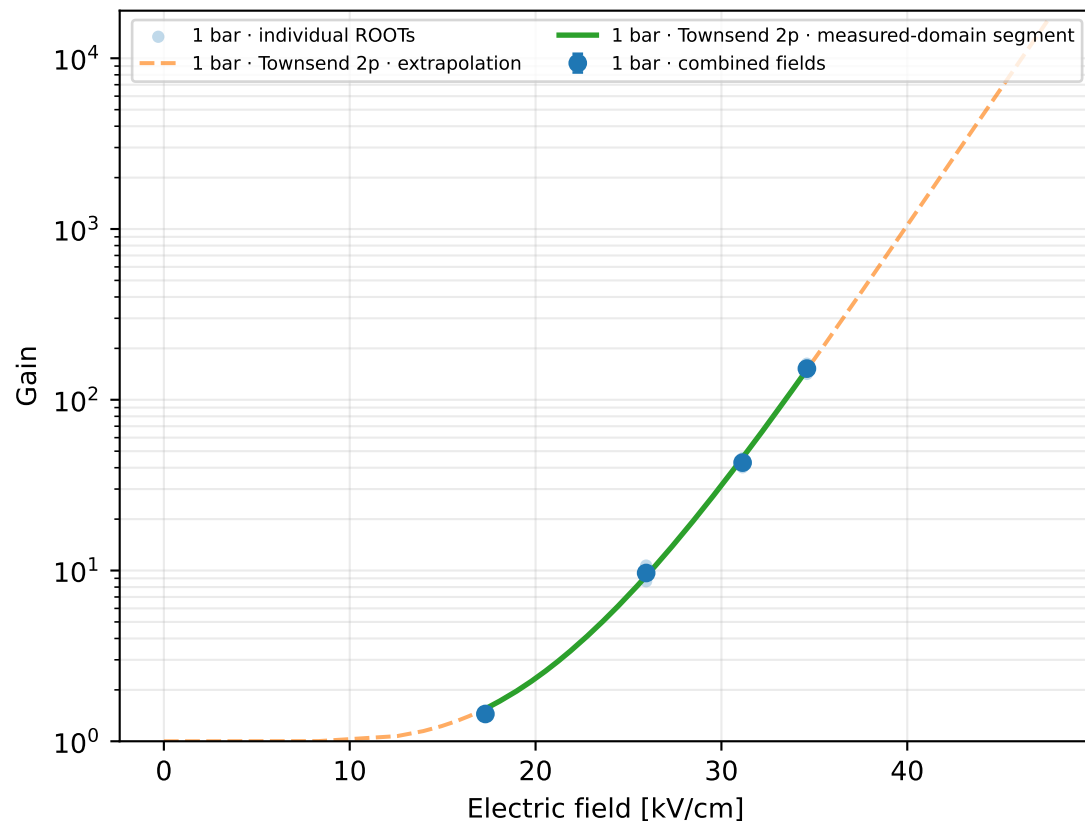
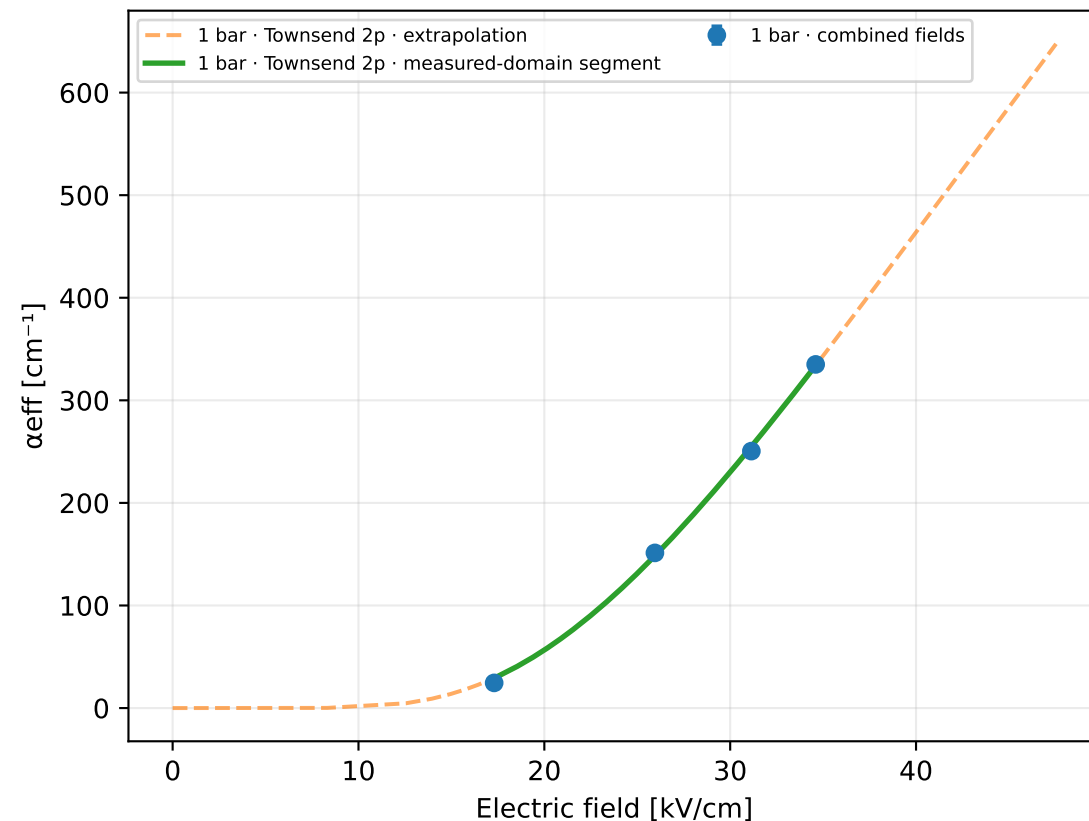


ArCF4Iso · Ar 87 % / CF₄ 10 % / Iso 3 % · gap 0.15 mm

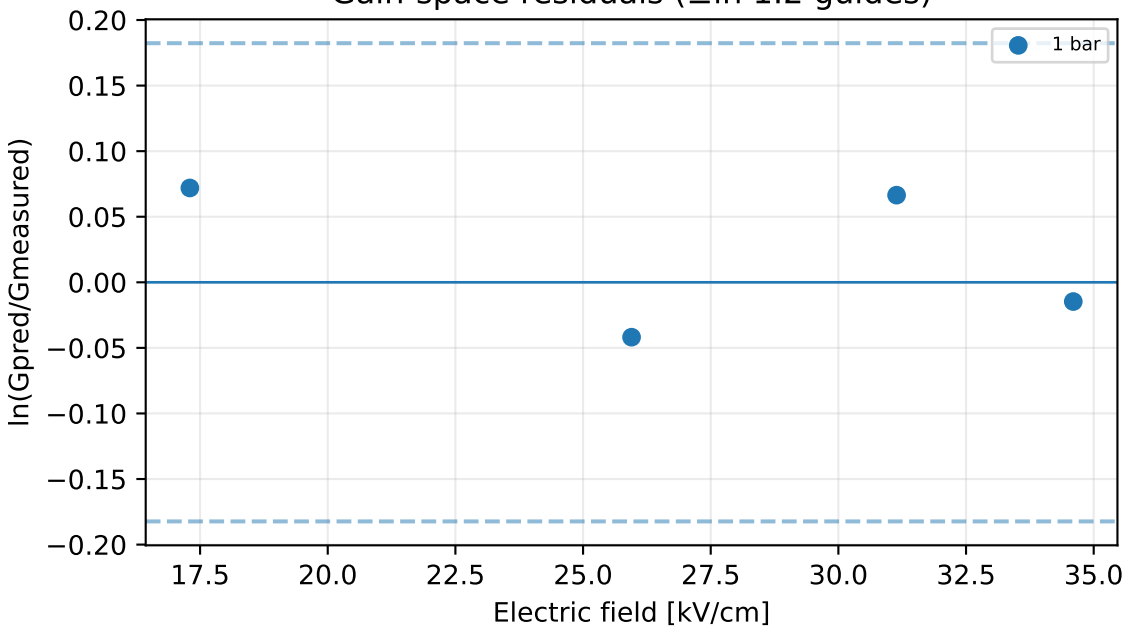
Gain vs electric field



Effective Townsend coefficient vs electric field



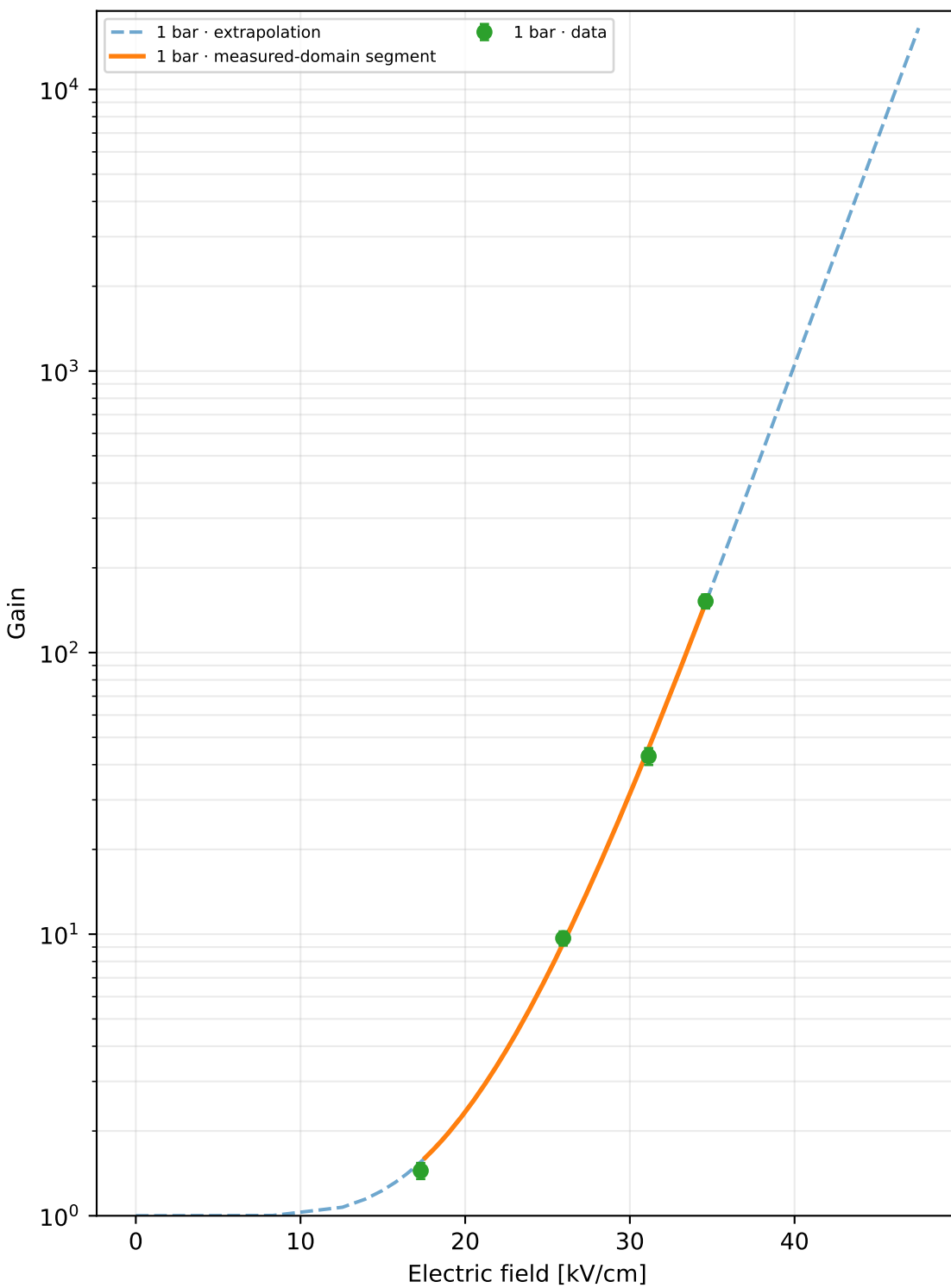
Gain-space residuals ($\pm \ln 1.2$ guides)



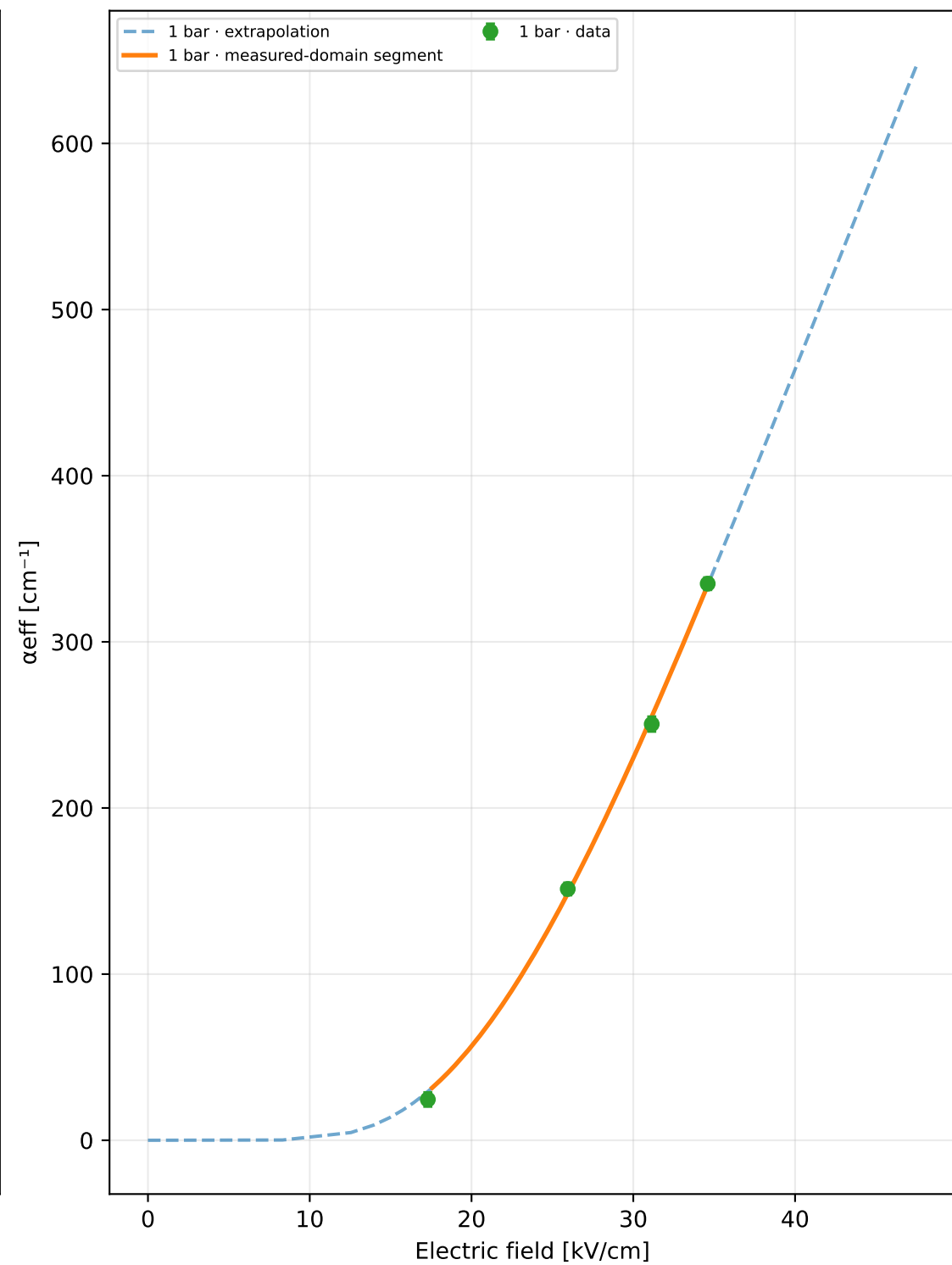
ROOT runs: 8
 Unique physical points: 4
 Pressures: 1 bar
 Selected model: Townsend 2p
 Status: rejected_poor_cross_validation
 Training RMSE: $\times 1.094$
 Cross-validation median: unavailable
 Cross-validation worst: unavailable
 Reduced-field range: 17.3–34.6 kV cm⁻¹ bar⁻¹
 A=3803.5, B=84.162, m=0, n=1
 Covariance condition: 569
 Warnings: poor_cross_validation, large_cross_validation_outlier
 1 bar display limit: G=1.65e+04 (local inversion limit)

Extrapolation capped at $G=1e+07$ or at the local model limit

Gain vs electric field



α_{eff} vs electric field



p [bar]	E [kV/cm]	E/p	Gain	σ Gain	α_{eff} [cm ⁻¹]	npe	runs
1	17.3	17.3	1.445	0.0945	24.5406	200	2
1	25.95	25.95	9.67	0.559	151.269	200	2
1	31.14	31.14	42.84	2.94	250.498	200	2
1	34.6	34.6	152.205	8.72	335.015	200	2

Progressive model selection

The higher-order model is selected only when new points improve out-of-sample prediction.

Candidate	k	AICc	CV median	CV worst	usable	warnings
Townsend 2p	2	-3.25	—	—	no	poor cross validation, large cross validation outlier